Companion technical note

Magnetic Sheaves and Stability Walls in a Compact Calabi–Yau Transition

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Abstract: We study magnetic sheaves in a compact Calabi–Yau transition coupling two conifold sectors to a rank-one E_2 sector. Fixed-support Hilbert schemes give rank-one numerical Donaldson–Thomas contributions on the rigid $\mathbb{F}_1$ and $\mathbb{P}^2$ surfaces on the two sides. On a reducible moving divisor, two dual-ideal sheaves are isolated and Gieseker stable, each contributing $+1$; their stability persists along ample polarizations approaching the contraction. An explicit magnetic splitting has one extension mode and a transverse phase alignment near large volume. Using the analytically continued periods of the companion paper, we determine the constituent phase sign at the marked quantum transition and exclude emission of the known light electric particles on a final specified segment. The probe charges nevertheless carry confined magnetic flux after Higgsing. These are component counts and partial wall constraints, not a determination of the physical BPS index at the endpoint. Counts involving complete moving pencils require a separate reduced-member hypothesis, stated in an appendix.

Keywords: D-branes, M-Theory, Supersymmetric Gauge Theory, String Duality

Contents

1 Geometry, charge conventions and scope

Magnetic objects can test aspects of a Calabi–Yau transition that are not fixed by its massless electric spectrum. Such a test requires both a charge calculation and a stability condition. This note studies the first requirement and parts of the second for the compact transition of [1]. We compute specified numerical Donaldson–Thomas contributions, their polarization dependence, a two-constituent magnetic wall near large volume, and exact phase constraints near the quantum transition. The results do not determine the full physical BPS index at the strongly coupled endpoint.

Let $X = \widehat{X}_9$ be the smooth resolution of a compact Calabi–Yau threefold with two conifold points and one cone over $E = \mathrm{Bl}_2\,\mathbb{P}^2$. Its Hodge numbers are $(6, 122)$. The nodal curves C_1, C_2 and the two exceptional curves e_1, e_2 of E obey

$$R = \iota_*(e_1 - e_2) = C_1 + C_2. \tag{1.1}$$

The compact gaugings allow these local sectors to Higgs only together. The smooth side has Hodge numbers $(3, 123)$ and a rigid plane. In the resolved chamber the corresponding surface is $S_- = D_8 \simeq \mathbb{F}_1$; on the smooth side it is $S_+ \simeq \mathbb{P}^2$, with class ν. Their classical restrictions and string anomalies are computed in [1, Section 3].

We use the primitive divisor basis

$$\mathcal{D} = (D_2, D_5, D_6, D_7, D_8, E). \tag{1.2}$$

The integral nef basis $(H_1, \dots, H_6)$ has columns

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 5 & 5 \\ 0 & 1 & 2 & 4 & 4 & 5 \\ 0 & -1 & -1 & -3 & -3 & -3 \\ 0 & -1 & -1 & -2 & -2 & -2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{pmatrix} \tag{1.3}$$

in $\mathcal{D}$. In particular $H_1 = D_5$, $H_2 = D_4 + D_5$, $H_3 = D_1$, $H_4 = D_0$. Write β_a for the dual curve basis. Then

$$C_2 = \beta_1, \quad C_1 = \beta_5, \quad \iota_* e_2 = \beta_6, \quad \iota_* e_1 = \beta_1 + \beta_5 + \beta_6. \tag{1.4}$$

The ample class $J_* = \sum_a H_a = (6, 18, 16, -11, -8, 7)$ fixes the initial polarization. The contraction face is $J = aD_0 + bD_1$, $a, b \geq 0$. Appendix B prints the intersection data needed for the sheaf stability tests.

For a pure surface sheaf $\mathcal{F}$, charges mean

$$\Gamma(\mathcal{F}) = \mathrm{ch}(\mathcal{F})\sqrt{\mathrm{Td}(X)} = (0, P, q, q_0), \qquad Z_{\mathrm{LV}} = -\int_X e^{-t}\Gamma, \quad t = B + iJ. \tag{1.5}$$

The D0 convention makes a point sheaf have period -1. The Dirac pairing with an electric charge $(0, 0, \gamma, \gamma_0)$ is $P\gamma$. The normalized exact central charge for $P \neq 0$ differs from its polynomial value by a magnetic period correction Ψ_P. No additional curvature term is added after the Mukai charge has been fixed.

The four-dimensional particles here come from wrapping a five-dimensional magnetic string on an auxiliary spacetime circle. This is distinct from the elliptic circle in the six-dimensional interpretation of X. The physical mass contains the usual common normalization; phase comparisons use the holomorphic central charges above.

The main unconditional sheaf result is the existence of two isolated stable components carrying relative charges $(C_2, -1)$ and $(R, -1)$ with respect to a moving D4 seed. Each contributes $+1$ to the numerical DT invariant. Both remain Gieseker stable along ample polarizations approaching the contraction. Their explicitly identified magnetic split has a transverse alignment near large volume, but the exact endpoint periods lie strictly away from that alignment. Light-electric emission walls are also excluded on a final specified segment. These exclusions are incomplete: they do not prove that the constituent BPS states survive the whole continuation. Moreover their magnetic charges are confined at a generic Higgs vacuum. Conditional counts involving the full moving pencils are kept in Appendix A so that their additional genericity assumption is separate from the isolated-component results.

2 Supported sheaves and isolated contributions

The rigid surfaces $S_- = \mathbb{F}_1$ and $S_+ = \mathbb{P}^2$ also give a computable component of a protected count. Compactify the spatial direction of the wrapped-M5 string on a circle. The resulting type-IIA charges are D4–D2–D0 charges. We use the large-volume description by Gieseker-stable pure two-dimensional sheaves, with an ample polarization on the compact threefold X, and count the component supported on S. This is a specified sheaf-moduli contribution, not an identification of the full physical BPS index at the transition point.

Fix a line bundle L on S and let $Z \subset S$ have length n. Write $i : S \hookrightarrow X$ and $\mathcal{E} = L \otimes I_Z$. Every rank-one torsion-free sheaf on S with determinant L and second Chern class n is of this form. Its pushforward is stable for every ample polarization: a proper subsheaf of the same rank has a nonzero lower-dimensional quotient and hence smaller reduced Hilbert polynomial. The moduli space on S is $S^{[n]} = \mathrm{Hilb}^n(S)$, smooth and projective of dimension $2n$ [2].

This remains a smooth component of the moduli on the compact threefold. The normal bundle is K_S, and the change-of-rings sequence gives

$$0 \longrightarrow \mathrm{Ext}^1_S(\mathcal{E}, \mathcal{E}) \longrightarrow \mathrm{Ext}^1_X(i_*\mathcal{E}, i_*\mathcal{E}) \longrightarrow \mathrm{Hom}_S(\mathcal{E}, \mathcal{E} \otimes K_S). \tag{2.1}$$

The last group vanishes: taking double duals embeds it into $H^0(S, K_S) = 0$. There are no determinant deformations, since $H^1(S, \mathcal{O}_S) = 0$. Thus the first group is the $2n$-dimensional tangent space to $S^{[n]}$. The natural map from this proper Hilbert scheme into the stable-sheaf moduli is therefore an isomorphism onto an open and closed smooth component. In particular, the ambient Behrend function is $(-1)^{2n} = 1$ on this component, so its numerical Donaldson–Thomas contribution is

$$\Omega_{S,L}(n) = \chi(S^{[n]}) \tag{2.2}$$

[3, Theorem 4.18 and Section 1.2]. No other moduli component or stability chamber is included in this definition.

The cohomology supplies a refinement $\Omega_{S,L}(n;y) = y^{-2n}\sum_j b_j(S^{[n]})y^j$. Here y records centered cohomological degree; it is not a flavor fugacity or an assumed infrared R-charge. We do not identify this refinement with a protected spin character at finite gravitational coupling: a spin-refined trace in the compact string theory need not be invariant under changes of coupling [4, Section 2.6]. For these two rational surfaces all odd Betti numbers vanish. Göttsche's product formula [2, eq. (2.1)] gives

$$\begin{aligned} \mathcal{Z}_S(q,y) &= \sum_{n\geq 0}\Omega_{S,L}(n;y)q^n \\ &= \prod_{m\geq 1}\frac{1}{(1-y^{-2}q^m)(1-q^m)^{b_2(S)}(1-y^2q^m)}. \end{aligned} \tag{2.3}$$

The answer is independent of L. The first numerical contributions are

n	0	1	2	3	4	5	6
S_-	1	4	14	40	105	252	574
S_+	1	3	9	22	51	108	221

(2.4)

Since $b_2(S_-) - b_2(S_+) = 1$, even the refined ratio is

$$\frac{\mathcal{Z}_{S_-}(q,y)}{\mathcal{Z}_{S_+}(q,y)} = \prod_{m\geq 1}(1-q^m)^{-1}. \tag{2.5}$$

With $q = e^{2\pi i\tau}$, the vacuum-normalized unrefined series $q^{-\chi(S)/24}\mathcal{Z}_S(q,1)$ are $\eta(\tau)^{-4}$ and $\eta(\tau)^{-3}$. Their ratio is the oscillator character $\eta(\tau)^{-1}$ of the extra chiral boson, consistent with the wrapped-string anomaly difference computed in [1]. These are fixed-flux coefficients of the rank-one string count [5, Section 2.1], not its complete flux-summed elliptic genus.

2.1 Shifted flux charges

The flux shift is essential when interpreting these coefficients as brane charges. In the convention $\Gamma = \mathrm{ch}(i_*\mathcal{E})\sqrt{\mathrm{Td}(X)}$, Grothendieck–Riemann–Roch gives

$$\begin{aligned} f &= c_1(L) - \frac{1}{2}K_S, \\ \Gamma &= \left(0, [S], i_*f, \frac{1}{2}\int_S f^2 + \frac{\chi(S)}{24} - n\right) \end{aligned} \tag{2.6}$$

[5, Appendix B.1]. Thus $c_1(L) = ah+be$ on S_- gives $f = (a+\frac{3}{2})h + (b-\frac{1}{2})e$, whereas $c_1(L) = ah$ on S_+ gives $f = (a+\frac{3}{2})h$. For the fixed representatives with restrictions $(D_0|, D_1|, D_8|) = (0, h, -3h+e)$ on S_- and $(D_0|, D_1|, \nu|) = (0, h, -3h)$ on S_+, their electric charges are

$$\begin{aligned} (Q_{D_0}, Q_{D_1}, Q_{D_8})_- &= (0, a+\tfrac{3}{2}, -3a-b-4), \\ (Q_{D_0}, Q_{D_1}, Q_{\nu})_+ &= (0, a+\tfrac{3}{2}, -3a-\tfrac{9}{2}). \end{aligned} \tag{2.7}$$

There is no integral b for which the shifted exceptional flux vanishes. Consequently (2.5) compares oscillator counts, not equal-charge states across the transition. A flux-summed genus requires the shifted lattice, a quadratic refinement, and its polarization-dependent self-dual decomposition. Nor does this sheaf calculation determine all bound states in the physical stability condition of the compactification.

2.2 Magnetic pairings

There is an additional obstruction to extracting the condensate spectrum from this string by ordinary halo wall-crossing. For a D4–D2–D0 core of magnetic charge P and a purely electric particle $\gamma = (0, 0, \gamma_2, \gamma_0)$, choose the Dirac pairing convention $\langle \Gamma_P, \gamma\rangle = P \cdot \gamma_2$. The compact data give

$$\begin{array}{c|ccr} P & C_1 & C_2 & R = C_1 + C_2 \\ \hline D_8 & 0 & 0 & 0 \\ D_5 & 0 & 1 & 1 \\ D_6 & -1 & 0 & -1 \end{array} \tag{2.8}$$

The first row vanishes for every electric charge in $\mathbb{Z}C_1 + \mathbb{Z}C_2$, independently of the core flux and of D0 charge. The ordinary halo index has an exponent proportional to this Dirac pairing [6]; see also [4, eqs. (2.23)–(2.25)]. Its multiplier around the D_8 core is identically one for any spectrum supported on this charge lattice. Equivalently, $D_5|_{S_-} = D_6|_{S_-} = 0$, and every supported flux counted above is neutral under both effective vectors. This excludes a halo-wall-crossing determination using this core; it does not exclude threshold bound states, contact interactions, or the geometric transition itself. The D_5 and D_6 strings instead have nonzero pairings with the required root and complementary nodes. This vanishing applies to the rank-two root and nodal lattice, not to all four finite-radius conifold charges: $D_8\iota_*e_1 = D_8\iota_*e_2 = 1$. The distinction will matter in Section 4.4.

2.3 Moving supports and the reducible fiber

The nonzero pairings in (2.8) do not by themselves determine a probe index. Unlike D_8, both D_5 and D_6 move. Write $X = \widehat{X}_9$ and $P_i = D_i$. Their pencils have the members

$$P_5 \sim D_3 + D_7 + D_8, \qquad P_6 \sim D_4 + D_7 + D_8, \qquad P_5^2 = 0, \quad P_6^2 = C_1. \tag{2.9}$$

The first defines the K3 fibration. A smooth member of the second is a K3 surface blown up at a point, with canonical curve C_1. Each pencil has two sections, represented in Cox coordinates by $(z_5, z_3z_7z_8)$ and $(z_6, z_4z_7z_8)$. The second has scheme-theoretic base locus C_1, a smooth complete intersection with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$. Its universal member is therefore $\mathrm{Bl}_{C_1} X$. Counting a single fixed K3 would omit the motion of the support [7, Section 2.2].

The higher cohomology of $\mathcal{O}_X(P_i)$ vanishes and $h^0(\mathcal{O}_X(P_i)) = 2$. The divisor sequence then gives $H^1(T, \mathcal{O}_T) = 0$ for every member. The structure-sheaf seed charges, whether or not their full pencil is a stable moduli component, are

$$\Gamma_{5,0} = (0, P_5, 0, 1), \qquad \Gamma_{6,0} = (0, P_6, -\tfrac{1}{2}C_1, \tfrac{11}{12}). \tag{2.10}$$

Appendix A computes zero- and one-point pencil contributions under a separate reduced-member hypothesis. The support restrictions and isolated contributions below are unconditional for the specified resolved geometry.

Where the missing probe states must occur. For P_5, the limitation is stronger than motion of the support. Any effective support cycle of class P_5 is vertical for ρ: nefness and $P_5^2 = 0$ imply that each component has zero intersection with P_5 and an ample divisor. A sheaf on such a support is unchanged by tensoring with $\mathcal{O}_X(P_5)$, which is pulled back from the base and trivial over a neighborhood of the finitely many supporting fibers. Its induced D2 charge therefore obeys

$$Q_{P_5} = P_5 \cdot \mathrm{ch}_2(\mathcal{F}) = 0. \tag{2.11}$$

Adding C_2 or R to a seed changes this charge by one. Such a core–particle state cannot lie in any pure two-dimensional sheaf component of magnetic charge P_5 in this large-volume description. It would require a different physical stability description, for example a derived object; this is not a no-go for the physical state.

For P_6, set $A_6 = D_7 - D_6 = -D_4 - D_8$. Every pencil member except $T_* = D_4 + D_7 + D_8$ is disjoint from D_4 and D_8. Thus $\mathcal{O}_X(A_6)$ restricts trivially to it, and all sheaves supported on those members have $Q_{A_6} = 0$, including the seeds. But

$$A_6 \cdot C_1 = 0, \qquad A_6 \cdot C_2 = A_6 \cdot R = 1. \tag{2.12}$$

Within this pencil, a sheaf obtained by adding C_2 or R must therefore be supported on T_*. This reduces the sheaf calculation to a reducible surface, with gluing along its double curves. The vertical modular formulas of [7] assume irreducible fibers and cannot be applied directly to this contribution. Neither a fixed smooth-K3 oscillator series nor the seed components above determine it.

An isolated contribution with cooperative electric charge. Isolated components of this reducible-surface moduli problem can be computed without the additional pencil hypothesis. Put $T = T_*$ and $\mathcal{I} = I_{C_2,T}$. Consider the two sheaves

$$\mathcal{F}_2 = \mathcal{H}om_T(\mathcal{I}, \mathcal{O}_T), \qquad \mathcal{F}_R = \mathcal{H}om_T(\mathcal{I}, \omega_T) = \mathcal{F}_2 \otimes \mathcal{O}_X(P_6). \tag{2.13}$$

Sheaves on T are again understood to be pushed forward to X. Both are J_*-Gieseker stable and define isolated reduced points of the stable-sheaf moduli on X. Their charges and numerical DT contributions are

$$\begin{array}{c|c|c} & \Gamma & \Omega^{\mathrm{iso}} \\ \hline \mathcal{F}_2 & (0, P_6, -\tfrac{1}{2}C_1 + C_2, -\tfrac{1}{12}) & 1 \\ \mathcal{F}_R & (0, P_6, \tfrac{1}{2}C_1 + C_2, -\tfrac{1}{12}) & 1. \end{array} \tag{2.14}$$

In particular, $\Gamma(\mathcal{F}_R) - \Gamma_{6,0} = (0, 0, R, -1)$. The superscript denotes these individual components, not the full invariants at their charges.

The construction also determines the gluing at the double curve. The complete intersection $C_2 = D_2 D_4 \simeq \mathbb{P}^1$ has normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ in X, and

$$(D_4, D_7, D_8) \cdot C_2 = (-1, 1, 0), \qquad P_6 \cdot C_2 = 0. \tag{2.15}$$

It meets $D_4 \cap D_7$ at one transverse point p, away from D_8. Near p the surface and ideal have the form $\mathcal{O}_T = \mathbb{C}[[x, y, z]]/(xy)$ and $\mathcal{I} = (x, z)$. Thus C_2 is not Cartier on T: the dual ideal in (2.13) is rank-one torsion-free, but not locally free at p. This is essential. The naive restrictions $(\mathcal{O}_{D_4}(C_2), \mathcal{O}_{D_7}, \mathcal{O}_{D_8})$ have degrees one and zero on the two sides of $D_4 \cap D_7$, and cannot be glued as a line bundle.

The ideal $\mathcal{I}$ is maximal Cohen–Macaulay on the Gorenstein surface T, by the sequence $0 \to \mathcal{I} \to \mathcal{O}_T \to \mathcal{O}_{C_2} \to 0$ and the depth lemma. Duality gives pure sheaves in (2.13) and sequences

$$\begin{aligned} &0 \longrightarrow \mathcal{O}_T \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{O}_{C_2}(-2) \longrightarrow 0, \\ &0 \longrightarrow \omega_T \longrightarrow \mathcal{F}_R \longrightarrow \mathcal{O}_{C_2}(-2) \longrightarrow 0. \end{aligned} \tag{2.16}$$

Here $\omega_T = \mathcal{O}_T(P_6) = \mathcal{O}_T(C_1)$, and its restriction to C_2 is trivial. Since $\chi(\mathcal{O}_{C_2}(-2)) = -1$, these sequences give precisely the D0 shift in (2.14); it must not be discarded when interpreting the electric flux as an added particle.

For stability, a proper subunion $U \subset T$ has the same generic double-curve gluing as $\mathcal{O}_T$. For $T = U + (T - U)$ define

$$\delta_T(U) = [J_* U(2T - U)](J_*^2 T) - (J_* T^2)(J_*^2 U).$$

Relative to $\mathcal{O}_T$, the component fluxes of $\mathcal{F}_2$ are $(C_2, 0, 0)$ and those of $\mathcal{F}_R$ are $(C_2, C_1, 0)$, in the order (D_4, D_7, D_8). Let γ be the total flux and γ_U its sum over U. The numerator comparing the maximal subsheaf supported on U with the full sheaf is

$$\delta_\gamma(U) = \delta_T(U) + 2\big[(J_* \cdot \gamma)(J_*^2 \cdot U) - (J_* \cdot \gamma_U)(J_*^2 \cdot T)\big]. \tag{2.17}$$

Point defects do not change this coefficient. For the six subunions $1, 2, 3, 12, 13, 23$, the numerators are

$$\begin{array}{c|cccccc} \mathcal{F}_2 & 3388 & 5222 & 1488 & 1482 & 4282 & 4334 \\ \mathcal{F}_R & 3630 & 4974 & 1494 & 1476 & 4530 & 4092. \end{array} \tag{2.18}$$

All are strictly positive. Full-multirank proper subsheaves have a nonzero lower-dimensional quotient, hence a smaller reduced Hilbert polynomial. This proves stability, including the non-locally-free gluing point.

To compute the ambient deformation space, let I_{C_2} be the ideal in X. Since C_2 is disjoint from the pencil base curve C_1, the restriction of the two sections of $\mathcal{O}_X(P_6)$ to $\mathcal{O}_{C_2}(P_6) = \mathcal{O}_{C_2}$ is surjective. Together with the higher cohomology vanishing above, this gives $R\,\mathrm{Hom}(\mathcal{O}_X(-P_6), I_{C_2}) = \mathbb{C}$ in degree zero. Consequently

$$0 \longrightarrow \mathcal{O}_X(-P_6) \longrightarrow I_{C_2} \longrightarrow \mathcal{I} \longrightarrow 0 \tag{2.19}$$

is the spherical-twist triangle. The autoequivalence of [8, Theorem 1.2] identifies the self-Ext groups of I_{C_2} and $\mathcal{I}$. The former has no first-order deformations: $H^0(N_{C_2/X}) = 0$ and $H^1(\mathcal{O}_X) = 0$. Derived duality satisfies $R\mathcal{H}om_X(\mathcal{I}, \mathcal{O}_X) = \mathcal{F}_R[-1]$; it, and tensoring by $\mathcal{O}_X(-P_6)$, preserve the dimension of self-Ext^1. Thus

$$\mathrm{Ext}^1_X(\mathcal{F}_2, \mathcal{F}_2) = \mathrm{Ext}^1_X(\mathcal{F}_R, \mathcal{F}_R) = 0. \tag{2.20}$$

Each stable point is therefore an open and closed smooth zero-dimensional component, with Behrend weight $+1$ [3, Section 1.2 and Theorem 4.18].

The second sheaf supplies a compact contribution carrying the cooperative electric class. It remains distinct from the desired five-dimensional particle index. The complete moduli at either charge have not been classified, and physical stability at the cooperative contraction is not established. Section 3 extends the Gieseker-stability calculation and explains why the raw D0 shift is frame-dependent. At these fixed charges, the divisor sequence and Serre duality give

$$R\,\mathrm{Hom}_X(\mathcal{O}_{C_2}(-1), \mathcal{O}_T) = 0.$$

Thus replacing the quotient in (2.16) by the zero-D0 curve sheaf does not give a nontrivial extension of $\mathcal{O}_T$. This excludes that elementary construction, not other sheaves or derived objects at zero D0 shift.

3 Stability and magnetic constituents

The two isolated sheaves remain stable for polarizations approaching the cooperative face. This statement concerns Gieseker stability; it does not assume its equivalence to physical stability at vanishing curve volume. Set

$$J_\epsilon = aD_0 + bD_1 + \epsilon J_*, \qquad a, b \geq 0, \quad \epsilon > 0. \tag{3.1}$$

This class is ample. For $a + b > 0$ it approaches the contraction as $\epsilon \to 0$, with $J_\epsilon \cdot C_1 = J_\epsilon \cdot C_2 = \epsilon$. Substitution in (2.17) gives cubics in (a, b, ϵ) with nonnegative coefficients for all six subunions, for both $\mathcal{F}_2$ and $\mathcal{F}_R$. The coefficient of ϵ^3 is the strictly positive entry in (2.18). Explicit common lower bounds are printed in (B.3). Thus neither sheaf crosses a Gieseker wall along (3.1); its isolated contribution remains $+1$ for every positive rational ϵ and rational $a, b \geq 0$.

The magnetic probe itself does not contract. Its limiting volume is

$$\lim_{\epsilon \to 0} \frac{1}{2} J_\epsilon^2 \cdot P_6 = 7a^2 + 8ab + 2b^2 > 0 \qquad (a + b > 0). \tag{3.2}$$

Moreover, Gieseker stability is an asymptotic statement about the Hilbert polynomial for each fixed ample polarization. A physical large-volume approximation along λJ_ϵ requires, in particular, $\lambda\epsilon \gg 1$ in string units. It is not uniform in the contraction limit at fixed λ. The calculation therefore removes a polarization-wall obstruction, but does not control quantum stability at the endpoint [5, Appendix B.2].

Circle momentum and spectral flow. There are two different circle reductions in this paper. The elliptic fibration describes a six-dimensional theory reduced to five dimensions; its KK tower is represented by M2-branes on the elliptic fiber class F. The sheaf count instead compactifies a spatial direction of the five-dimensional M5 string on an auxiliary circle to obtain a four-dimensional D4–D2–D0 particle. D0 charge records momentum on this latter circle [5, Section 2.1]. It must not be identified with the elliptic-fiber contribution in the six-dimensional dictionary.

The relative D0 label also depends on the integral B-field frame. For a line-bundle class $\ell \in H^2(X,\mathbb{Z})$, tensoring gives, in our Mukai convention,

$$\begin{aligned} (P,q,q_0) &\longmapsto (P, q+P\ell, q_0+\ell\cdot q+\tfrac{1}{2}P\ell^2), \\ (\gamma,\Delta q_0) &\longmapsto (\gamma, \Delta q_0+\ell\cdot\gamma) \end{aligned} \tag{3.3}$$

for two charges with the same magnetic part P. The simultaneous change $B\mapsto B+\ell$ leaves $e^{-B}\Gamma$ unchanged; this is a change of charge frame, not a new physical state. It is the line-bundle form of spectral flow [5, eq. (2.7)], with the D0 sign fixed by our definition of Γ. Here

$$P_6D_5 = F, \qquad D_5^2 = 0, \qquad D_5\cdot C_2 = D_5\cdot R = 1. \tag{3.4}$$

Consequently $\ell = D_5$ changes the relative labels $(C_2,-1)$ and $(R,-1)$ to $(C_2,0)$ and $(R,0)$, while adding F to the seed's D2 charge. A nonzero raw D0 shift is therefore not an invariant obstruction. This relabeling neither supplies a state at the original seed-plus-root charge in the original $B=0$ frame nor extracts a five-dimensional particle from a wrapped magnetic string. Decompactifying the auxiliary circle also removes the finite momentum spacing, but makes the wrapped magnetic particle an extended string; its excitation count is not thereby a bulk particle index.

A fixed-charge path crosses a magnetic slope wall. Varying B while keeping the sheaves and their charges fixed is different from the simultaneous transformation above. Take $B=-sD_5$, $0\le s\le 1$, and define the twisted Hilbert polynomial by $P_{B,J}(\mathcal{F};m) = \int_X e^{-B}\,\mathrm{ch}(\mathcal{F})e^{mJ}\,\mathrm{Td}(X)$. For a subunion U, its strict linear-coefficient comparison becomes

$$\delta_{\mathcal{F},s}(U) = \delta_{\mathcal{F},0}(U) + 2s\big[(J\cdot F)(J^2\cdot U) - (J\cdot D_5U)(J^2\cdot T)\big], \qquad T = P_6. \tag{3.5}$$

All the flux $D_5T = F$ lies on D_4. The first vanishing comparison is therefore the one for $U = D_4$, at

$$s_{\mathcal{F}}(J) = \frac{\delta_{\mathcal{F},0}(D_4)}{2(J\cdot F)\,J^2\cdot(D_7+D_8)} \in (0,1). \tag{3.6}$$

To check that this is first, multiply each of the other five comparisons at this value by the positive denominator. Their expansions in (a,b,ϵ) have nonnegative coefficients and strictly positive ϵ^6 coefficients. Also $\delta_{\mathcal{F},1}(D_4)$ has strictly negative coefficients on every nonzero monomial. The intersections printed in Appendix B and (3.5) determine these polynomial tests. In particular,

$$\begin{aligned} s_{\mathcal{F}_2}(J_*) &= \frac{77}{104}, & s_{\mathcal{F}_R}(J_*) &= \frac{165}{208}, \\ \lim_{\epsilon\to 0} s_{\mathcal{F}}(J_\epsilon) &= \frac{7a^2+8ab+2b^2}{11a^2+10ab+2b^2}. \end{aligned} \tag{3.7}$$

The limiting value is 17/23 on the interior ray $a = b$. Both sheaves are twisted-Gieseker stable before their stated slope wall and unstable after it. At the wall itself the constant Hilbert coefficient must also be compared; no claim of Gieseker semistability there is needed.

The destabilizing subsheaf and the extension are explicit. Write $V = D_7 + D_8$ and $\mathcal{G} = \mathcal{O}_{D_4}(C_2 + K_{D_4})$. The component gluing gives

$$\begin{aligned} &0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{O}_V \longrightarrow 0, \\ &0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{F}_R \longrightarrow \mathcal{O}_V(P_6) \longrightarrow 0. \end{aligned} \tag{3.8}$$

These are nontrivial extensions, unique up to scalar. Indeed $H^*(\mathcal{G}) = 0$ and $H^*(\mathcal{G}(V)) = \mathbb{C}$ in degree zero: on the rational elliptic surface D_4, the two line bundles are $\mathcal{O}(C_2 - F)$ and $\mathcal{O}(C_2)$. The divisor resolution of $\mathcal{O}_V$ then gives $\mathrm{Ext}^1_X(\mathcal{O}_V, \mathcal{G}) = \mathbb{C}$ and vanishing in other degrees. The same holds for $\mathcal{O}_V(P_6)$ because $\mathcal{O}_{D_4}(P_6)$ is trivial. The two magnetic charges are D_4 and V, with Dirac pairing -1 in our convention. This is a split of the magnetic support, not emission of a purely electric cooperative particle.

For comparison, the polynomial large-volume central charge $Z_{\mathrm{LV}} = -\int_X e^{-(B+iJ)}\Gamma$ [6, Appendix A, eq. (A.9)] gives relative values

$$\Delta Z_{2,\mathrm{LV}} = 1 - s + i\epsilon, \qquad \Delta Z_{R,\mathrm{LV}} = 1 - s + 2i\epsilon. \tag{3.9}$$

The fixed-charge path thus encounters the magnetic slope wall before the real term cancels at $s = 1$. This is a controlled large-volume comparison, not an exact marginal-stability wall at the contraction. We next determine the associated two-constituent contribution near large volume and the persistence of its phase alignment.

3.1 Magnetic bound states near large volume

The magnetic splitting in (3.8) admits a two-constituent bound-state calculation near large volume. It also gives a transverse phase alignment that persists under sufficiently small quantum corrections. Neither statement requires extrapolating Gieseker stability to the contraction.

Rigid constituents and one extension mode. Write $A = D_4$, $V = D_7 + D_8$, $\mathcal{H}_2 = \mathcal{O}_V$ and $\mathcal{H}_R = \mathcal{O}_V(T)$. All three constituents $\mathcal{G}, \mathcal{H}_2, \mathcal{H}_R$ are spherical: their self-Ext groups are $\mathbb{C}$ in degrees zero and three and vanish in degrees one and two. For $\mathcal{G}$, this follows from $H^1(A, \mathcal{O}_A) = H^0(A, K_A) = 0$ on the rational elliptic surface. For either extension $0 \to \mathcal{G} \to \mathcal{F} \to \mathcal{H} \to 0$, the cross-Ext calculation above and Serre duality give

$$\mathrm{RHom}_X(\mathcal{H}, \mathcal{G}) = \mathbb{C}[-1], \qquad \mathrm{RHom}_X(\mathcal{G}, \mathcal{H}) = \mathbb{C}[-2]. \tag{3.10}$$

The connecting map $\mathrm{Ext}^2(\mathcal{G},\mathcal{H}) \to \mathrm{Ext}^3(\mathcal{G},\mathcal{G})$ is an isomorphism: its Serre-dual map sends the identity of $\mathcal{G}$ to the nonzero extension class. Thus $\mathrm{RHom}(\mathcal{G},\mathcal{F}) = \mathbb{C}$ in degree zero, and $\mathcal{H}$ is the spherical twist of $\mathcal{F}$ about $\mathcal{G}$. The autoequivalence [8, Theorem 1.2] and the rigidity of $\mathcal{F}$ prove the assertion for $\mathcal{H}$.

The factors are also twisted-Gieseker stable along the polarizations (3.1) and $B = -sD_5$. Stability of $\mathcal{G}$ follows from rank one on the integral surface A. For $\mathcal{H} = \mathcal{O}_V(L)$, where $L = 0$ or T, the maximal subsheaf on $U = D_7$ or D_8 is $\mathcal{O}_U(L-(V-U))$. Put $q_U = U(L-(V-U)) - U^2/2$. Its strict slope comparison is

$$(J_\epsilon \cdot q_{\mathcal{H}})(J_\epsilon^2 \cdot U) - (J_\epsilon \cdot q_U)(J_\epsilon^2 \cdot V) > 0. \tag{3.11}$$

Each of these four cubics has nonnegative coefficients in (a,b,ϵ); their ϵ^3 coefficients, in the order $(\mathcal{H}_2, D_7), (\mathcal{H}_2, D_8), (\mathcal{H}_R, D_7), (\mathcal{H}_R, D_8)$, are $(422, 282, 419, 285)$. These inequalities are independent of s because $D_5V = 0$. Proper subsheaves of full multirank have a nonzero lower-dimensional quotient and cannot destabilize.

Phase alignment including the D0 terms. Fix J_ϵ and use $t = -sD_5 + i\lambda J_\epsilon$. Define $v_A = J_\epsilon^2 A/2$, $v_V = J_\epsilon^2 V/2$, $f = J_\epsilon \cdot F$, $g = J_\epsilon \cdot q_{\mathcal{G}}$ and $h = J_\epsilon \cdot q_{\mathcal{H}}$. Their values are

$$\begin{aligned} v_A &= \tfrac{3}{2}a^2 + 3ab + b^2 + 21a\epsilon + 17b\epsilon + \tfrac{121}{2}\epsilon^2, \\ v_V &= \tfrac{11}{2}a^2 + 5ab + b^2 + 45a\epsilon + 19b\epsilon + 88\epsilon^2, \\ f &= 3a + 2b + 13\epsilon, \qquad g = -\tfrac{1}{2}(3a + 2b + 11\epsilon), \\ h_2 &= \tfrac{1}{2}(3a + 2b + 12\epsilon), \qquad h_R = \tfrac{1}{2}(3a + 2b + 14\epsilon). \end{aligned} \tag{3.12}$$

The polynomial central charges, with the Mukai D0 terms retained, are therefore

$$Z_{\mathcal{G},\mathrm{LV}} = \lambda^2 v_A - s + \tfrac{1}{2} + i\lambda(g + sf), \qquad Z_{\mathcal{H},\mathrm{LV}} = \lambda^2 v_V - \tfrac{5}{12} + i\lambda h. \tag{3.13}$$

Set $K = fv_V > 0$ and $C = v_A h - v_V g$. Direct multiplication gives

$$\begin{aligned} \mathrm{Im}(Z_{\mathcal{G},\mathrm{LV}}\overline{Z_{\mathcal{H},\mathrm{LV}}}) &= \lambda\big[(\lambda^2 K + h - \tfrac{5}{12}f)s - \lambda^2 C - \tfrac{5}{12}g - \tfrac{1}{2}h\big], \\ s_{\mathcal{F}}^{\mathrm{pol}}(\lambda) &= \frac{\lambda^2 C + \frac{5}{12}g + \frac{1}{2}h}{\lambda^2 K + h - \frac{5}{12}f}. \end{aligned} \tag{3.14}$$

Here $4C = \delta_{\mathcal{F},0}(A)$ and $4K$ is the denominator in (3.6). Hence this alignment approaches the slope wall with an $O(\lambda^{-2})$ correction. At J_*,

$$s_2^{\mathrm{pol}}(\lambda) = \frac{20328\lambda^2 + 17}{27456\lambda^2 + 14}, \qquad s_R^{\mathrm{pol}}(\lambda) = \frac{21780\lambda^2 + 29}{27456\lambda^2 + 38}. \tag{3.15}$$

Both lie in $(0,1)$ for $\lambda \geq 1$. Their derivatives of the alignment function with respect to s are strictly positive. The real parts of both constituent central charges are positive there, so the solutions describe equal phases, not opposite phases. The same conclusions hold for any fixed J_ϵ and sufficiently large λ.

Quantum corrections and the local bound-state index. Choose flat period coordinates $t^I = X^I/X^0$ in the large-volume charge frame, and divide the holomorphic central charge by X^0. For zero D6 charge its form is

$$Z(0,P,q,q_0;t) = -\tfrac{1}{2}Pt^2 + q\cdot t - q_0 + \Psi_P(t), \tag{3.16}$$

where Ψ_P is linear in P and contains the worldsheet-instanton corrections to the magnetic periods [5, Appendix B.2]. The polynomial charge convention is the one used in (3.13); no additional c_2 shift is added to its D0 terms. Specify the path by $t = -sD_5 + i\lambda J_\epsilon$ in these flat coordinates. For fixed ample J_ϵ, the convergent large-volume period expansion makes Ψ_A, Ψ_V and their s derivatives $O(e^{-c\lambda})$ for some $c > 0$, uniformly on a bounded s interval.

The correction to the first line of (3.14) is then $O(\lambda^2 e^{-c\lambda})$, whereas its transverse derivative is $\lambda^3 K + O(\lambda)$. The implicit-function argument gives a unique nearby exact alignment, with

$$s_{\mathcal{F}}^{\text{exact}}(\lambda) = s_{\mathcal{F}}^{\text{pol}}(\lambda) + O(\lambda^{-1}e^{-c\lambda}). \tag{3.17}$$

The same-phase inequality remains strict, since $\mathrm{Re}(Z_{\mathcal{G}}\overline{Z_{\mathcal{H}}}) = \lambda^4 v_A v_V + O(\lambda^2) > 0$. This proves persistence of the alignment near large volume without evaluating the quantum periods. The estimate is not uniform at the contraction.

Under the usual large-volume identification of these stable sheaves with BPS branes, the two-constituent sector near alignment has one chiral extension mode x, no adjoint deformation modes and no oriented cycle. Its quiver quantum mechanics has dimension vector $(1,1)$ and no superpotential. With the sign convention $\zeta \propto -\mathrm{Im}(Z_{\mathcal{G}}\overline{Z_{\mathcal{H}}})$, its relative D-term equation is

$$|x|^2 = \zeta, \qquad \mathcal{M}_{\text{rel}} = \begin{cases} \{\text{pt}\}, & \zeta > 0, \\ \varnothing, & \zeta < 0. \end{cases} \tag{3.18}$$

At $\zeta = 0$ the constituents can separate and the bound state is at threshold. The relative index is consequently one below the nearby wall in s and zero above it. Equivalently the two-constituent contribution to the index jump, oriented from the unbound to the bound side, is $+1$: both isolated constituent contributions are $+1$ and $|\langle\Gamma_{\mathcal{G}}, \Gamma_{\mathcal{H}}\rangle| = 1$. This agrees with the primitive wall-crossing factor [6, Section 4.2 and eq. (5.3)]. It is a contribution from specified rigid constituents, not the complete invariant of either constituent charge or of their sum.

One quantum-period identity needs no expansion. Since the charged sheaf and the seed have the same magnetic charge, their magnetic periods cancel exactly:

$$\Delta Z_2 = 1 + t\cdot C_2, \qquad \Delta Z_R = 1 + t\cdot R. \tag{3.19}$$

These equal $1 - s + i\lambda\epsilon$ and $1 - s + 2i\lambda\epsilon$ on the specified flat-coordinate path. The identity does not assert that a BPS particle of the difference charge exists, or identify the classical contraction with a particular quantum-period value. In particular, the bounds above are for fixed $\epsilon > 0$ and are not uniform as $\epsilon \to 0$ at fixed λ. Section 4 evaluates their endpoint phase sign using the continued periods of the companion paper.

4 Quantum phase constraints

Mirror symmetry supplies closed-M2 indices independently of the magnetic sheaf counts. Let n^0_β be the genus-zero Gopakumar–Vafa invariant of the resolved threefold in curve class β. We fix its normalization by

$$\mathcal{F}_{\text{inst}}(t) = -\frac{1}{(2\pi i)^3} \sum_{\beta>0} n^0_\beta \,\mathrm{Li}_3(e^{2\pi i t\cdot\beta}). \tag{4.1}$$

These are integer coefficients of a spin-weighted M2 index, not dimensions of magnetic sheaf moduli [9]. Applying the toric Frobenius construction [10, 11] gives

$$n^0_{C_1} = n^0_{C_2} = 1, \qquad n^0_R = 0, \qquad n^0_{\iota_* e_1} = n^0_{\iota_* e_2} = 1, \tag{4.2}$$

where $\iota : E \hookrightarrow \widehat{X}_9$ and $e_1 = D_4|_E$, $e_2 = D_3|_E$. The period construction and integral marking are given in [1, Section 5 and Appendix C]. It determines all coefficients through total nef degree eight: of 3002 nonzero lattice classes in the chosen outer Mori cone, 80 have nonzero genus-zero invariant.

The mixed electric class has no connected curve. The vanishing in (4.2) also has a geometric proof, valid beyond the finite degree computation. Let $\pi : \widehat{X}_9 \to X_9$ be the contraction, whose exceptional fibers are the disjoint loci C_1, C_2 and E. For any $m, n > 0$, a connected stable map in class $\beta = mC_1 + nC_2$ would have zero degree against the pullback of an ample divisor on X_9. Its image must therefore lie in one fiber of π. It cannot lie in either nodal fiber, since C_1, C_2 are independent classes and both coefficients are positive. It cannot lie in E:

$$E\cdot\beta = 0, \qquad E|_E = K_E, \qquad -K_E \text{ is ample.} \tag{4.3}$$

Every nonconstant effective curve in E has strictly negative intersection with E. Hence the connected stable-map space is empty for this class in every genus. In particular $n^0_{mC_1+nC_2} = 0$ for all $m, n > 0$. This argument concerns connected maps; the disconnected cycle $C_1 \sqcup C_2$ certainly exists.

The other entries of (4.2) are the isolated $(-1,-1)$ rational-curve contributions, reproduced by the mirror calculation. The compact relation now has a spectral interpretation:

$$\iota_* e_1 = \iota_* e_2 + C_1 + C_2, \qquad J\cdot\iota_* e_1 = J\cdot\iota_* e_2 + J\cdot C_1 + J\cdot C_2. \tag{4.4}$$

An isolated rational curve in E has the same total compact charge and five-dimensional BPS mass as the disconnected three-curve configuration on the right. The genus-zero single-particle index at the left-hand charge is $+1$, whereas the difference class R has no connected holomorphic representative. This is a threshold charge and mass identity, not a calculation of a decay amplitude. It explains why the $+1$ magnetic contribution at relative charge $(R, -1)$ cannot be read as an elementary particle of charge R. Neither the vanishing nor this threshold identity determines the interacting operator spectrum at the singular point.

Computed magnetic corrections. Let $H_1, \dots, H_6$ be the integral nef basis specified in Section 1, and let β_a be its dual curve basis. Thus $\beta_1 = C_2$, $\beta_5 = C_1$, $\beta_6 = \iota_* e_2$, and $\iota_* e_1 = \beta_1 + \beta_5 + \beta_6$. Write $t = \sum_a t_a H_a$ and $q_a = e^{2\pi i t_a}$. Differentiating (4.1) gives the corrections in (3.16) in the same Mukai frame:

$$\Psi_P(t) = \frac{1}{4\pi^2} \sum_{\beta > 0} (P \cdot \beta) n^0_\beta \operatorname{Li}_2(q^\beta). \tag{4.5}$$

No additional polynomial or c_2 term is included here. Since $A = -H_1 + H_2$ and $V = H_1 - H_2 - H_5 + H_6$, the first terms are

$$\begin{aligned} 4\pi^2 \Psi_A &= -q_1 + q_2 + \tfrac{1}{4}(-q_1^2 + q_2^2) + 3q_2q_3 - 2q_2q_6 + O(q^3), \\ 4\pi^2 \Psi_V &= q_1 - q_2 - q_5 + q_6 + \tfrac{1}{4}(q_1^2 - q_2^2 - q_5^2 + q_6^2) - 3q_2q_3 + q_3q_6 + O(q^3). \end{aligned} \tag{4.6}$$

The remainder denotes total degree at least three, not powers of a single coordinate. The one-quarter coefficients are double covers of the degree-one classes.

To evaluate the previous path, use the flat coordinates

$$(t_1, \dots, t_6) = (-s + i\lambda\epsilon,\ i\lambda\epsilon,\ i\lambda(b+\epsilon),\ i\lambda(a+\epsilon),\ i\lambda\epsilon,\ i\lambda\epsilon). \tag{4.7}$$

Let $s_{[d]}$ denote the root of the alignment equation after retaining GV terms of total degree at most d in (4.5), with the full dilogarithm for each retained class. At $t = -sD_5 + iJ_*$ the degree-eight values are

	s^{pol}	$s_{[8]}$	$s_{[8]} - s^{\mathrm{pol}}$
$\mathcal{F}_2$	0.740626137605	0.740632681385	6.54378×10^{-6}
$\mathcal{F}_R$	0.793227613297	0.793233852071	6.23877×10^{-6}

(4.8)

The degree-six and degree-eight results differ by less than 2×10^{-20} here. This is agreement between truncations, not a rigorous bound on the omitted classes. The truncated roots are transverse and satisfy the same-phase inequality.

These numbers compare finite truncations near large volume. They are not used to infer the phase sign at the transition.

4.1 Charge transport and circle holonomy

There is an exact continuation statement near either ordinary conifold, with the other sectors kept nonsingular. Put $\tau_i = t \cdot C_i$ and choose the vanishing brane $\mathcal{O}_{C_i}(-1)$, whose charge is $(0, 0, C_i, 0)$. The isolated-curve contribution is

$$\Psi_P^{(i)} = \frac{P \cdot C_i}{4\pi^2} \operatorname{Li}_2(e^{2\pi i \tau_i}) = \frac{P \cdot C_i}{2\pi i} \tau_i \log(-2\pi i \tau_i) + \text{single-valued terms}. \tag{4.9}$$

A counterclockwise loop around $\tau_i = 0$ consequently adds $(P \cdot C_i)\tau_i$ to Z. In the original period branch this is equivalent to the integral charge transformation

$$M_i : (P, q, q_0) \longmapsto (P, q + (P \cdot C_i)C_i, q_0). \tag{4.10}$$

We use this period-continuation convention; transforming the charge basis instead gives the inverse map. This is the usual conifold transvection, also described by the spherical-twist monodromy [12, Section 5].

The two transformations commute, since the vanishing charges are mutually local. They are not equivalent to a single conifold transvection associated with R. On the integral magnetic–electric lattice,

$$\begin{aligned}(M_1M_2-1)(P,q) &= \big(0,(P\cdot C_1)C_1+(P\cdot C_2)C_2\big),\\ (M_R-1)(P,q) &= \big(0,(P\cdot R)R\big),\\ \operatorname{rank}(M_1M_2-1)=2,&\qquad \operatorname{rank}(M_R-1)=1.\end{aligned} \tag{4.11}$$

Here M_R is a hypothetical single-particle transvection, not an asserted discriminant monodromy. This distinction is consistent with $n^0_R=0$ and is not removed by the compact homology relation. For the magnetic constituents the needed pairings are

$$\begin{array}{l|rr} P & P\cdot C_1 & P\cdot C_2\\ \hline A & 0 & -1\\ V & -1 & 1\\ T & -1 & 0\end{array}. \tag{4.12}$$

Thus M_2 changes the electric charges of $\mathcal{G}$ and $\mathcal{H}$ by $-C_2$ and $+C_2$, respectively, while leaving their total probe charge fixed. The combined nodal monodromy shifts the probe and its seed by $-C_1$, rather than $-R$. Their relative charges remain unchanged. These formulas determine charge transport, not stability or an index at the common singular point.

On (4.7), the normalized electric periods of the two nodal D0 towers are

$$Z_{1,k}=k+i\lambda\epsilon,\qquad Z_{2,k}=k-s+i\lambda\epsilon,\qquad k\in\mathbb{Z}. \tag{4.13}$$

At fixed auxiliary-circle radius, simultaneous nodal masslessness therefore requires $\epsilon\to 0$ and $s\to\mathbb{Z}$. If the alignment tends to an interior value $s_*\in(0,1)$ instead, the second tower has a nonzero normalized mass gap $\operatorname{dist}(s_*,\mathbb{Z})$. Thus shrinking the classical curve areas along this B-field path does not by itself reach the simultaneous massless locus. This is a condition on the auxiliary four-dimensional probe problem, not an obstruction to the original five-dimensional geometric transition.

4.2 The marked compact transition

Write $(v,a,b)=(z_2,z_3,z_4)$ for the spectator parameters. For sufficiently small nonzero values, the companion paper proves that the locus

$$\Sigma:\quad z_1=z_5=z_6=1\quad\Longleftrightarrow\quad t_1=t_5=t_6=0 \tag{4.14}$$

has four ordinary nodes. In the continued large-volume marking, their integral charges are precisely

$$(0,0,C_1,0),\quad (0,0,C_2,0),\quad (0,0,\iota_*e_1,0),\quad (0,0,\iota_*e_2,0). \tag{4.15}$$

They generate a primitive rank-three lattice, with the single relation $\iota_* e_1 - \iota_* e_2 - C_1 - C_2 = 0$. This assertion uses convergent electric periods and unit local residues, not just the abstract relation or Hodge numbers [1, Section 5.2 and Appendix C]. Its analytic proof is not repeated here.

In this compact integral marking, the magnetic pairings with the exceptional-curve doublet are

$$\begin{array}{c|rr} P & P\cdot\iota_* e_1 & P\cdot\iota_* e_2 \\ \hline A & -1 & 0 \\ V & 1 & 1 \\ T & 0 & 1 \end{array}. \tag{4.16}$$

Thus the monodromy around this pair shifts the electric charge of $\mathcal{G}$ by $-\iota_* e_1$, that of $\mathcal{H}$ by $\iota_* e_1 + \iota_* e_2$, and the total probe and its seed each by $\iota_* e_2$. Their relative charges remain unchanged. With $\tau_i = Z_{\gamma_i}$, the corresponding singular part of a magnetic period due to these two nodes is

$$Z_P^{\mathrm{sing}} = \frac{1}{2\pi i}\sum_{i=1}^{2}(P\cdot\iota_* e_i)\tau_i \log\tau_i. \tag{4.17}$$

The two C_1, C_2 nodes contribute the analogous terms with their previously computed magnetic pairings. Each singular term tends to zero at the massless point and does not determine the regular, bulk-dependent part of either constituent period. In particular it does not fix $\mathrm{Im}(Z_{\mathcal{G}}\overline{Z_{\mathcal{H}}})$. These singular pieces alone do not settle the phase alignment. The regular terms can, however, be continued on the same small-bulk branch.

4.3 Regular periods and the constituent phase

The same continuation gives the regular magnetic periods [1, Appendix C]. In the Mukai frame of (3.16), their leading terms are

$$\begin{aligned} \Psi_A|_\Sigma &= -\frac{1}{12} - \frac{v+a}{4\pi^2} + O(\delta^2), \\ \Psi_V|_\Sigma &= \frac{1}{12} + \frac{2(v+a)}{4\pi^2} + O(\delta^2), \qquad \delta = |v|+|a|+|b|. \end{aligned} \tag{4.18}$$

Here Σ denotes (4.14); the error is a small-bulk analytic remainder, not a total nef-degree truncation. The constant terms include the four conifold multicovers. The convergence proof in the companion paper sums all local degrees before expanding in the spectator parameters.

For positive sufficiently small (v, a, b), all three spectator flat coordinates are purely imaginary. Write $t_j = iy_j$ for $j = 2, 3, 4$, with $y_j > 0$, and put $J_\Sigma = y_2 H_2 + y_3 H_3 + y_4 H_4$. The restricted magnetic corrections are real. The two quotient charges differ by C_1 with zero D0 shift, so their periods coincide exactly on Σ. Using their Mukai charges gives

$$\begin{aligned} Z_{\mathcal{G}} &= \mathcal{V}_A + \tfrac{1}{2} + \Psi_A - iY, \\ Z_{\mathcal{H}_2} = Z_{\mathcal{H}_R} &= \mathcal{V}_V - \tfrac{5}{12} + \Psi_V + iY, \quad Y = y_3 + \tfrac{3}{2}y_4, \end{aligned} \tag{4.19}$$

where $\mathcal{V}_P = PJ_\Sigma^2/2$. In particular, the common total probe period is real and equals

$$\begin{aligned} M_\Sigma &= \mathcal{V}_T + \tfrac{1}{12} + \Psi_T, & \Psi_T &= \Psi_A + \Psi_V, \\ \mathcal{V}_T &= 2y_2y_3 + 2y_3^2 + 3y_2y_4 + 8y_3y_4 + 7y_4^2. \end{aligned} \tag{4.20}$$

It is strictly positive when the positive bulk parameters are sufficiently small: $y_3, y_4 \to +\infty$, whereas $\Psi_T \to 0$. For either specified constituent split the exact alignment function therefore satisfies

$$\operatorname{Im}(Z_{\mathcal{G}}\overline{Z_{\mathcal{H}}}) = -YM_\Sigma < 0. \tag{4.21}$$

Thus the target is not on this marginal-stability wall. With $\langle \Gamma_{\mathcal{G}}, \Gamma_{\mathcal{H}} \rangle = -1$, the necessary two-center stability inequality is satisfied [6, eq. (3.23)]; equivalently the relative Fayet–Iliopoulos parameter has the bound-state sign used in (3.18). This conclusion is a phase test for the specified charges. It does not prove survival of both constituent BPS states from large volume, exclude other decay channels, or extend the two-node quiver index to a locus with four additional massless particles. Those questions remain necessary for a compact magnetic bound-state index.

4.4 Light-particle emission and confinement

Fix a positive sufficiently small bulk point on Σ. Approach it in flat coordinates with the spectator coordinates fixed and

$$t_1 = i\epsilon c_2, \qquad t_5 = i\epsilon c_1, \qquad t_6 = i\epsilon c_e, \qquad c_1, c_2, c_e > 0, \quad \epsilon \downarrow 0. \tag{4.22}$$

The four light periods, ordered as $(C_1, C_2, \iota_* e_1, \iota_* e_2)$, are exactly $i\epsilon(c_1, c_2, c_1 + c_2 + c_e, c_e)$. At the endpoint, (4.19) gives positive real parts for $Z_{\mathcal{G}}$ and both $Z_{\mathcal{H}}$ when the bulk parameters are sufficiently small. The same holds for both total probe periods. Continuity, including the vanishing $\tau \log \tau$ terms, preserves these inequalities on a final segment $0 < \epsilon < \epsilon_0$.

Let Γ be any of these five charges and let $\eta = \sum_i n_i \gamma_i$, with $n_i \geq 0$ and not all zero, be a sum of the four light charges. Then $Z_\eta = i\epsilon m_\eta$, where $m_\eta = n_1c_1 + n_2c_2 + n_3(c_1 + c_2 + c_e) + n_4c_e > 0$, and

$$\operatorname{Im}(Z_{\Gamma-\eta}\overline{Z_\eta}) = -\epsilon m_\eta \operatorname{Re} Z_\Gamma < 0. \tag{4.23}$$

Thus $\Gamma \to (\Gamma - \eta) + \eta$ cannot be marginal on this segment. The CPT-conjugate electric charge has the opposite, also nonzero, alignment. This excludes these light-emission walls, whether or not the proposed residual charge supports a BPS state. It neither excludes other magnetic decompositions nor supplies a stable state at the beginning of the segment. At $\epsilon = 0$ the electric masses vanish, so this argument does not define an endpoint particle index.

Confined magnetic flux. Choose the orientations

$$(\delta_1, \delta_2, \delta_3, \delta_4) = (-\gamma_{C_1}, -\gamma_{C_2}, \gamma_{e_1}, -\gamma_{e_2}), \qquad \sum_i \delta_i = 0. \tag{4.24}$$

At a generic Higgs vacuum all four hypermultiplets have nonzero expectation values. A D4 magnetic charge $P \in H^2(\widehat{X}_9, \mathbb{Z})$ then has flux winding vector $w(P) = (\langle P, \delta_i \rangle)_{i=1}^4$, where the pairing is $P \cdot \gamma_i$ before the orientation change. Its image lies in

$$\Lambda_{\text{flux}} = \{w \in \mathbb{Z}^4 : \textstyle\sum_i w_i = 0\}. \tag{4.25}$$

This is the A_3 root lattice as an integral winding lattice; it does not assert an $SU(4)$ gauge enhancement. Write $P = \sum_a p^a H_a$ in the primitive nef basis. The independent pairings are $(P \cdot C_1, P \cdot C_2, P \cdot \iota_* e_2) = (p^5, p^1, p^6)$. They range independently over $\mathbb{Z}^3$, so the image is all of (4.25), with no finite index. The matter charges generate a saturated lattice; consequently the unbroken subgroup of the $U(1)^6$ gauge torus is connected. The $\mathbb{Z}_4$ quotient of the Higgs-branch coordinates is not an unbroken discrete gauge group at a generic vacuum.

The relevant magnetic pairings are

$$\begin{array}{c|cccc|cccc} & PC_1 & PC_2 & P\iota_*e_1 & P\iota_*e_2 & w_1 & w_2 & w_3 & w_4 \\ \hline A & 0 & -1 & -1 & 0 & 0 & 1 & -1 & 0 \\ V & -1 & 1 & 1 & 1 & 1 & -1 & 1 & -1 \\ T & -1 & 0 & 0 & 1 & 1 & 0 & 0 & -1 \\ D_8 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & -1 \\ E & 0 & 0 & -1 & -1 & 0 & 0 & -1 & 1 \end{array}. \tag{4.26}$$

Here $A = D_4$, $V = D_7 + D_8$ and $T = D_6 = A + V$, as above. The fluxes of the two constituents do not cancel: their sum is $w(T) = (1, 0, 0, -1)$. Thus neither constituent nor either total probe can continue as an isolated particle with that magnetic charge at a generic Higgs vacuum. Its broken magnetic flux must be carried by strings. This is the usual confinement mechanism in a conifold Higgs transition [13, Sections 3–4]. Adding any electric charge leaves this obstruction unchanged. In the mirror surgery, the same pairings specify the boundary of the three-chain obtained from a magnetic three-cycle. Neither the existence of a particular BPS core before Higgsing nor the tension or BPS character of its flux strings follows from this charge calculation.

5 Interpretation and unresolved stability questions

The magnetic calculation gives three distinct types of result. The fixed-support Hilbert schemes and the isolated dual-ideal sheaves give numerical DT contributions in a specified large-volume description. The magnetic extension gives a one-mode bound-state contribution near its controlled large-volume wall. The analytically continued periods give exact phase signs near the quantum transition. Passing from one statement to another requires stability information; none is a substitute for the others.

In particular the isolated $+1$ contribution at relative charge $(R, -1)$ does not establish an elementary particle of charge R. The latter class has no connected holomorphic curve representative in this resolved chamber. Spectral flow can remove the relative D0 label only while transforming the core and B-field frame as well. The exact phase inequalities exclude the specified magnetic split and light-particle emissions near the endpoint, but not every possible decay. After Higgsing, the nonzero winding of the probe magnetic charge requires flux strings rather than an isolated particle.

To obtain a physical magnetic spectrum, one would need a stability condition compatible with the continued integral periods and control of the other objects at the relevant charges. Even at large volume, the complete moduli components have not been classified. The additional reduced-pencil hypothesis in Appendix A is another, independent geometric question. Until these problems are resolved, the appropriate conclusion is the set of component counts and wall constraints proved here, not a compact condensate index.

A Conditional pencil contributions

For this calculation impose the additional condition that every member of both pencils is reduced and that the displayed toric members are their only reducible members. This is an explicit hypothesis on the pencils, not a consequence of the cubic form; a complete genericity check for the anticanonical family is not supplied here. The isolated sheaves of Section 2.3 do not depend on this hypothesis.

For $n = 0, 1$, consider the sheaves $\mathcal{O}_T$ and $I_{p,T}$, respectively, on members $T \in |P_i|$, pushed forward to X. Use the ample polarization $J_* = (6, 18, 16, -11, -8, 7)$ in the basis (1.2); it is the sum of the six primitive ambient nef generators. These sheaves form open and closed smooth components of the stable pure-sheaf moduli, with contributions

$$\begin{array}{c|cc|cc} & \mathcal{M}_i(0) & \mathcal{M}_i(1) & \Omega_i^{\mathrm{seed}}(0) & \Omega_i^{\mathrm{seed}}(1) \\ \hline i=5 & \mathbb{P}^1 & X & -2 & 232 \\ i=6 & \mathbb{P}^1 & \mathrm{Bl}_{C_1} X & -2 & 230. \end{array} \tag{A.1}$$

The superscript denotes only these specified components, not the full index at the same charge.

Here is a component and stability argument, including the reducible members. For $T = A + B$, a proper subunion A supports the maximal subsheaf $\mathcal{O}_A(-B) \subset \mathcal{O}_T$. A subsheaf of $I_{p,T}$ supported on A has no larger reduced Hilbert polynomial. The strict comparison of the linear coefficients with that of $\mathcal{O}_T$ is

$$\delta_T(A) = [J_* \cdot A(A + 2B)](J_*^2 \cdot T) - (J_* \cdot T^2)(J_*^2 \cdot A) > 0. \tag{A.2}$$

For the ordered components in (2.9), and subunions $1, 2, 3, 12, 13, 23$, the numerators are

$$\begin{array}{c|rrrrrr} & 1 & 2 & 3 & 12 & 13 & 23 \\ \hline P_5 & 2952 & 3690 & 1230 & 1230 & 3690 & 2952 \\ P_6 & 3740 & 4876 & 1482 & 1488 & 4628 & 3982. \end{array} \tag{A.3}$$

Subsheaves of full multirank have a nonzero lower-dimensional quotient. The remaining pencil members are integral by hypothesis, where rank-one stability is automatic. Thus all the specified sheaves are stable, including those with the point on a double curve.

The smooth member determines the line-bundle cohomology: $h^0(X, \mathcal{O}_X(P_i)) = 2$, with no higher cohomology. The divisor sequence gives $H^1(T, \mathcal{O}_T) = 0$ for every member. Consequently $\operatorname{Ext}^1_X(\mathcal{O}_T, \mathcal{O}_T)$ is the one-dimensional pencil tangent space. For a point p outside the base locus, the evaluation map on the two sections is surjective, so $R\operatorname{Hom}(\mathcal{O}_X(-P_i), I_p) = \mathbb{C}$ in degree zero. The sequence

$$0 \longrightarrow \mathcal{O}_X(-P_i) \longrightarrow I_p \longrightarrow I_{p,T} \longrightarrow 0 \tag{A.4}$$

identifies $I_{p,T}$ with the spherical twist of I_p about $\mathcal{O}_X(-P_i)$. This twist is an autoequivalence [8, Theorem 1.2]; it preserves the three-dimensional $\operatorname{Ext}^1(I_p, I_p)$. For P_6 and $p \in C_1$, every pencil member is smooth along the base curve. The normal-deformation sequence then bounds $\dim \operatorname{Ext}^1_X(I_{p,T}, I_{p,T})$ by $2 + h^0(K_T) = 3$. The smooth three-dimensional incidence variety $\operatorname{Bl}_{C_1} X$ already supplies these deformations. Thus the natural proper parameter spaces are open and closed smooth components also at the base locus and singular fibers.

Their Behrend functions are -1. Since $\chi(X) = 2(6 - 122) = -232$ and $\chi(\operatorname{Bl}_{C_1} X) = \chi(X) + \chi(C_1) = -230$, the entries in (A.1) follow. In the charge convention of (2.6), the two families have

$$\Gamma_{5,n} = (0, P_5, 0, 1 - n), \qquad \Gamma_{6,n} = (0, P_6, -\tfrac{1}{2}C_1, \tfrac{11}{12} - n). \tag{A.5}$$

These results fix neither the other components nor the physical stability condition near the cooperative contraction.

B Intersection and stability data

The resolved fan and its primitive lattices are specified in [1, Appendix B]. In the ordered basis $\mathcal{D}$, with $D_9 = E$, the nonzero cubic entries are

IJK	κ_{IJK}	IJK	κ_{IJK}	IJK	κ_{IJK}	IJK	κ_{IJK}
222	8	266	1	667	-1	788	-6
225	-2	267	1	677	-1	888	8
226	-3	569	1	699	-2	889	-1
227	-2	599	-2	777	-3	899	-1
256	1	666	-1	778	4	999	7

(B.1)

All permutations agree and omitted entries vanish. The second-Chern pairings in this basis are

$$c_2(X) \cdot \mathcal{D} = (-4, 24, 26, 18, -4, -2).$$

The remaining divisors are

$$\begin{aligned} D_0 &= D_2 + 4D_5 + 4D_6 - 3D_7 - 2D_8 + 2E, \\ D_1 &= D_2 + 2D_5 + 2D_6 - D_7 - D_8 + E, \\ D_3 &= D_5 - D_7 - D_8, \qquad D_4 = D_6 - D_7 - D_8. \end{aligned} \tag{B.2}$$

The two nodal curves pair with $\mathcal{D}$ as $C_1 = (1, 0, -1, -1, 0, 0)$ and $C_2 = (-1, 1, 0, 1, 0, 0)$.

For the stability argument of Section 3, let Q_U have the coefficients below in the ordered monomials $(a^3, a^2b, ab^2, b^3, a^2\epsilon, ab\epsilon, b^2\epsilon, a\epsilon^2, b\epsilon^2, \epsilon^3)$. The subunion order is that of (2.18).

U	a^3	a^2b	ab^2	b^3	$a^2\epsilon$	$ab\epsilon$	$b^2\epsilon$	$a\epsilon^2$	$b\epsilon^2$	ϵ^3
1	42	76	44	8	553	662	190	2385	1420	3388
2	42	90	60	12	639	882	279	3141	2073	4974
3	0	42	48	12	70	476	237	660	1255	1488
12	0	42	48	12	70	476	233	660	1239	1476
13	42	90	60	12	595	842	275	2781	1937	4282
23	42	76	44	8	597	702	198	2745	1572	4092

(B.3)

For both sheaves, every coefficient of $\delta_{\mathcal{F},0}(U) - Q_U$ is nonnegative. Since Q_U has a positive ϵ^3 coefficient, all six comparisons are strictly positive on (3.1). The cubic data above also give the B-dependent comparisons directly through (3.5).

The exact Sage checks in `checks/` reconstruct the dual-ideal presentation, polarization polynomials, first slope walls and constituent phase data. A separate Python calculation checks the Hilbert-scheme products. The commands, including the mirror-period and numerical-wall calculations, are documented in `checks/README.md`. Finite-degree agreement does not certify a tail bound or a physical stability chamber.

References

[1] B.J. Wuebben, "Bulk gauging and quantum couplings across a compact higgs transition." 2026.

[2] L. Göttsche, "Hilbert schemes of points on surfaces." arXiv:math/0304302, 2003.

[3] K. Behrend, *Donaldson–Thomas type invariants via microlocal geometry*, *Ann. Math.* **170** (2009) 1307 [`math/0507523`].

[4] J. Manschot, B. Pioline and A. Sen, *Wall-crossing from Boltzmann black hole halos*, *JHEP* **07** (2011) 059 [`1011.1258`].

[5] M. Alim, B. Haghighat, M. Hecht, A. Klemm, M. Rauch and T. Wotschke, "Wall-crossing holomorphic anomaly and mock modularity of multiple M5-branes." arXiv:1012.1608, 2010.

[6] F. Denef and G.W. Moore, *Split states, entropy enigmas, holes and halos*, *JHEP* **11** (2011) 129 [`hep-th/0702146`].

[7] V. Bouchard, T. Creutzig, D.-E. Diaconescu, C. Doran, C. Quigley and A. Sheshmani, *Vertical D4–D2–D0 bound states on K3 fibrations and modularity*, *Commun. Math. Phys.* **350** (2017) 1069 [`1601.04030`].

[8] P. Seidel and R.P. Thomas, *Braid group actions on derived categories of coherent sheaves*, *Duke Math. J.* **108** (2001) 37 [`math/0001043`].

[9] R. Gopakumar and C. Vafa, "M-Theory and topological strings–II." arXiv:hep-th/9812127, 1998.

[10] S. Hosono, A. Klemm, S. Theisen and S.-T. Yau, *Mirror symmetry, mirror map and applications to Calabi–Yau hypersurfaces*, *Commun. Math. Phys.* **167** (1995) 301 [`hep-th/9308122`].

[11] M. Demirtas, M. Kim, L. McAllister, J. Moritz and A. Rios-Tascon, *Computational mirror symmetry*, *JHEP* **01** (2024) 184 [2303.00757].

[12] P.S. Aspinwall, M.R. Plesser and K. Wang, "Mirror symmetry and discriminants." arXiv:1702.04661, 2017.

[13] B.R. Greene, D.R. Morrison and C. Vafa, *A geometric realization of confinement*, *Nucl. Phys. B* **481** (1996) 513 [hep-th/9608039].